

VISHWAKARMA INSTITUTE OF INFORMATION TECHNOLOGY

RESEARCH INTERNSHIP REPORT ON

Information Extraction on Bio, Sports, Healthcare-Sports Domain with NER and Classification



Submitted by
Kaustubh Avinash Chaudhari
Student GR NO. 223M0001
(S.Y. M.Tech Computer)

Under the Guidance of
Dr. Manisha Mali

Department of Computer Engineering
Vishwakarma Institute of Information Technology, Pune - 48
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VISHWAKARMA INSTITUTE OF INFORMATION TECHNOLOGY

CERTIFICATE

This is to certify that

Kaustubh Avinash Chaudhari

(S.Y. M.Tech Computer)

Has successfully completed his project report on
Information Extraction on Bio, Sports, Healthcare-Sports
Domain with NER And Classification

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Dr. Manisha Mali
Guide

Dr. Sachin Sakhare
H.O.D. Computer Engineering

Dr. Vivek Deshpande
Director

Acknowledgement

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Kaustubh Avinash Chaudhari
Student GR NO. 223M0001
(S.Y. M.Tech Computer)

ABSTRACT

Due to the growing intersection between sports and healthcare, a large amount of domain-specific text data needs to be accurately classified and have detailed entity recognition performed. This paper outlines a holistic framework in Natural Language Processing (NLP) to automate domain classification and Named Entity Recognition (NER) in three crucial domains : biomedical, sports, and healthcare-sports. Within a BERT-based architecture, the system classifies sentences according to the relevant domain and extracts critical entities such as symptoms, injuries, diseases, drugs, tests, and sports-specific metrics with high performance. For enhancing searching ability at the entity level, each domain is paired with a customizable CSV-based dataset that allows for the relatively quick Extraction of relevant information via a ranked entity-matching algorithm. Our pipeline incorporates domain-specific NER models trained on synthetically generated, carefully curated BIO-tagged datasets, ensuring contextual richness and label consistency. Evaluation results show that extraction accuracy is very high across domains and is notably performance for multi-word medical and sports-healthcare terms

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1. Introduction

1.1 Introduction

In recent years, the convergence of healthcare and sports domains has posed unique challenges in information extraction due to the overlap of medical and athletic terminologies. Athletes, coaches, medical professionals, and analysts often rely on large volumes of unstructured text data—ranging from match reports to medical assessments—for insights into player conditions, injury trends, mental health, and drug-related violations. Extracting relevant and accurate information from such text requires not only domain-specific understanding but also precise categorization and named entity recognition (NER) capabilities. This project aims to address this critical need through an integrated NLP-based pipeline designed to classify domain-specific sentences and extract named entities with high precision across three distinct but related domains: sports, biomedical, and healthcare-sports.

The core contribution of the project is the creation and training of specialized NER models, particularly focused on the complex healthcare-sports domain, where symptoms, injuries, doping events, and psychological conditions often coexist in a single context. To achieve this, we curated custom datasets with thousands of annotated sentences generated using a wide range of manually designed templates. These templates were crafted to reflect real-world structures and language patterns found in medical-sports journalism, athlete interviews, doping reports, and clinical observations. Each sentence was labelled in BIO format with a focused set of entities such as INJURY, SYMPTOM, DRUG, MENTAL_HEALTH, DOPING_TEST, and SPORT_CONTEXT.

In addition to entity extraction, we implemented a BERT-based domain classifier to categorize sentences into the appropriate domain before applying the respective NER model. This dual-step approach ensures that contextually correct models are used for extraction, enhancing overall precision. The extracted entities are then used as search queries against pre-indexed CSV datasets from each domain, allowing the system to retrieve and rank the most relevant existing texts based on the number of matched entities.

This project demonstrates the practicality and scalability of custom entity-based NLP systems in highly contextual multi-domain scenarios. Through a structured pipeline involving text classification, BIO-tagged entity recognition, and partial-matching-based Extraction, it bridges the gap between synthetic dataset training and real-world querying performance. The final output not only extracts and categorizes information but also presents users with the most contextually relevant records from domain-specific databases—making it a powerful tool for professionals working at the intersection of sports and healthcare.

1.2 Scope

The scope of this project encompasses the design and development of a modular, domain-adaptive natural language processing (NLP) pipeline that can accurately process, classify, extract, and retrieve information from unstructured text. With the increasing volume of domain-blended data in healthcare and sports, the project aims to bridge the gap between general-purpose language models and specialized, context-aware information extraction systems. This solution is particularly relevant for scenarios where user-generated text contains mixed references to medical terminologies, sports events, athlete conditions, or healthcare-related cases in athletic settings.

The system is designed to work across three overlapping domains—Biomedical, Sports, and Healthcare-Sports—by employing pre-trained transformer models such as BERT for classification and token-level entity recognition. The pipeline can intelligently classify each sentence into its correct domain and route it through a domain-specific NER model that identifies key entities such as diseases, symptoms, drugs, players, injuries, doping references, and more. These entities are then used to query structured domain-specific datasets and return relevant information in a ranked format based on contextual overlap.

A unique aspect of the project lies in its support for sub-tag detection within the Healthcare-Sports domain, where certain terms may carry dual meanings across sports and medical contexts. The pipeline is capable of identifying subcategories such as *injury*, *mental health*, or *doping*, thereby enhancing the specificity and usefulness of retrieved information.

The system supports real-time query handling and delivers structured results without requiring large-scale backend databases, making it lightweight and adaptable. Its modular design ensures that the pipeline can be extended to additional domains or adapted to new datasets with minimal structural changes. As such, the scope of the project not only addresses current domain-specific information extraction challenges but also opens possibilities for future applications in cross-disciplinary NLP use cases, intelligent search systems, and automated report generation in professional settings such as sports medicine, clinical analysis, and athletic performance.

1.3 Objectives

- To effectively classify input sentences into one of the corresponding domains : the Bio domain, the Sports domain, or the Healthcare-Sports domain, applying a domain classifier within the framework of the present invention.
- To apply NERs that will act in a domain-specific manner to extract relevant entities per classified domain.

- To extract relevant entries from the corresponding domain-specific dataset based on those relevant entities and display that information.
- To detect and highlight relevant sub-category tags such as mental health, injury, or doping, as applicable within the Healthcare- Sports domain.

2. Literature Review

Recommendations algorithms are having difficulty handling sparsity, for example, healthcare wearable, where a user is least expected to have extensive interaction. To remedy that, a Privacy-Preserving Cross-Domain Healthcare Wearable Recommendation (PPCDHWRec) has been introduced. The technique brings in line how the interventions from users of different domains, such as E-commerce websites, can boost performance in healthcare wearable. It models how the user features can exist either in domain-dependent or domain-independent types by treating Funk-SVD and Orthogonal Nonnegative Matrix Tri-Factorization (ONMTF). The features are then fused together with a Factorization Machine to offer personalized recommendations while maintaining privacy. By obfuscating the item latent factors during transformation, PPCDHWRec achieves privacy protection without an accuracy sacrifice. Experimental results certify model accuracy and security for cross-domain recommendations in both well and poorly correlated domains [3].

To overcome the drawbacks of previous methods, which based their diagnosis on either time-domain or frequency-domain signals only, some more comprehensive techniques start to emerge from research: The Dual-domain Signal Fusion Adversarial Network (DSFAN), for instance, uses the Vision Transformer to extract and fuse features from time and frequency domains into a more integrated presentation of the operational conditions of bearings. As a further line of improving cross-domain diagnostic capability, the model also includes adversarial training based on Wasserstein distance, as a result of which the extracted features are expected to be domain-invariant. In fact, empirical evidence from the evaluation of DSFAN over six different cross-domain fault diagnosis tasks demonstrates its significant superiority over all previous methods in terms of both diagnostic accuracy and performance in domain adaptation, thus making this the most promising solution for actual industrial monitoring [1].

Biomedical information extraction mandates great precision owing to its specialized terminologies and unstructured complex text. In this regard, Named Entity Recognition (NER) is of utmost importance as it deals with the extraction of biological entities such as DNA, proteins, and RNA. The vast advancements in the pre-trained language models have begun to trump NER performance in almost all domains. This study selects and evaluates some of the modern state-of-the-art models—BERT-base, DistilBERT, ALBERT, XML-RoBERTa, ERNIE 2.0, and ConvBERT—for structured NER tasks on the BioNLP2004 dataset, which covers quite a vast array of biomedical entities. On comparing their performances, the research underlines the major challenges posed by biomedical texts and discusses how different fine-tuning strategies impact performance. The authors concluded with further evidence stressing the advantages of domain-specific pre-trained models for biomedical NER, firmly stating that they hold promise in pushing the envelope in the field of bioinformatics toward the more correct and efficient processing of biomedical knowledge[6].

Cross-domain sentiment classification is the task of classifying the sentiments expressed in texts concerning a certain category relevant to different kinds of classification of domain dependencies. Features of source and target domains mismatched from each other, which affect performance and degrade the overall accuracy of classification when it generates new or unseen target domains. Hence, a novel approach has been proposed to enhance an ESST (Enhanced Sentiment Sensitive Thesaurus) for bridging vocabulary gaps. The thesaurus matches the terms bearing sentiment across the domains and hence contributes towards generalization during the structural training of classifiers. In doing that, add terms to features vectors that are aligned with sentiments; hence, the process improves the overall performance of sentiment classifiers in new target domains. Empirical analysis on benchmark datasets, such as Amazon product reviews, confirms that the system improves target sentiment classification at the end of varying product categories, thus demonstrating the capacity of the ESST-enhanced cross-domain adaptation strategy-shown methods [2].

Data verity, reflected in the rare user-item interactions, has proved a critical threat to the prediction accuracy of recommender systems. The concept of cross-domain recommendation has come with a partial solution by transferring the useful knowledge from closely related domains to improve its predicted performance. It is within the already mentioned background that an article published recently ventured into research into anime and manga as similar domains with respect to their use for assessing cross-domain recommendation. Such research findings show that while the prediction error for using anime as an additional domain for manga has dropped to 0.075 in RMSE, and very high accuracy (>90%) has been sustained during extremes of sparsity (0.01%), the reverse - applying manga for anime recommendations - does not provide fairly similar gains. This asymmetry showcases the importance of directionality and correlation while working across domains. Domain-specific auxiliary knowledge is confirmed by the study to be helpful in enhancing recommendation performance under sparse countries [4].

The high speed with which web content grows in this information age makes the need for automated and efficient webpages' information extraction more crucial. There are impractical traditional manual approaches for locating the target resources in the overflow of information. This study proposes a page traversal algorithm that is intended to dynamically navigate and harvest information from web pages. Coupled with the tuning algorithms for the noise reduction processes, this method is expected to enhance accuracy, speed, and personalization of information Extraction. The scalable tool, especially for real time customers searching for a well-targeted content, will surely found an importance in web mining as well in intelligent information system development [9].

Nonetheless, the texts found in tourism-related travel blogs often contain a plethora of rich, unstructured information regarding various destinations and are usually very time-consuming to extract manually. To this end, the model proposed herein is a Named Entity Recognition (NER) model using BERT-BiLSTM-CRF to automatically identify and classify tourist attractions from textual narratives. The model uses the contextual embeddings of BERT, the sequential modelling offered by the BiLSTM, and the structured label assignment of CRF to extract natural landmarks, heritage sites, and man-made attractions. Trained on a public tourism NER dataset, the model managed to yield better performance on precision, recall, and F1 scoring measures of 77.2%, 83.04%, and 80.01%, respectively, putting the model ahead of baseline models. This shows how effective deep learning-based NER in retrieving information in the tourism domain can prove to be. [5].

Though the domain stays focused on some low-resource domains needing just little amounts of labelled data and an adversarial setup for localization, there have seen some ascents upon some of these systems in recent days. The BERT-NER model exploits the BERT architecture (Bidirectional Encoder Representations from Transformers) for enhanced entity recognition in different domains. A token classification layer, added above the BERT encoder and trained by cross-entropy loss, allows the model to account for contextual dependencies. Experiments with CoNLL-2003 dataset show that it attains very high scores of 93.12% precision, 94.46% recall, 93.79% F1-score, and 98.51% accuracy, thereby being a strong candidate of the NER systems of today. Transfer learning also makes it highly adaptable and robust for entity recognition solutions scalable across different domains [7].

Information extraction (IE) includes a very large unstructured data set and is concerned with identifying and extracting relevant facts in order to convert the facts to formal structured format. The paper deals with the evolution of IE techniques in great detail, with a special focus on the selected sub-tasks such as Named Entity Recognition (NER), relationship extraction, event extraction, and open information extraction (OpenIE). It details a variety of datasets and also gives real-world applications where these approaches will apply. Mapping the research trend in IE serves as a primary presentation to understand how structured knowledge can be generated systematically from textual content, for progressive strides in knowledge discovery, natural language understanding, and domain-specific analytics [8].

Continued expansion of web content has made it increasingly challenging to carry out efficient information Extraction in the modern-day information era. While users try to simple locate related resources among an overwhelming amount of data, there is an apparent rising thirst for accurate, automated, and personalized information-extraction techniques. A proposed solution and thus an answer to this daunting problem was the design and development of a web page traversal algorithm for identifying and extracting target

information. Additionally, tuning algorithms were incorporated to minimize interference and increase the clarity and relevance of the retrieved information. Basically, the technique was aimed at achieving real-time, high efficient processing of web information, ultimately providing an enhanced user experience in fast-changing online environments [10].

3. System Architecture

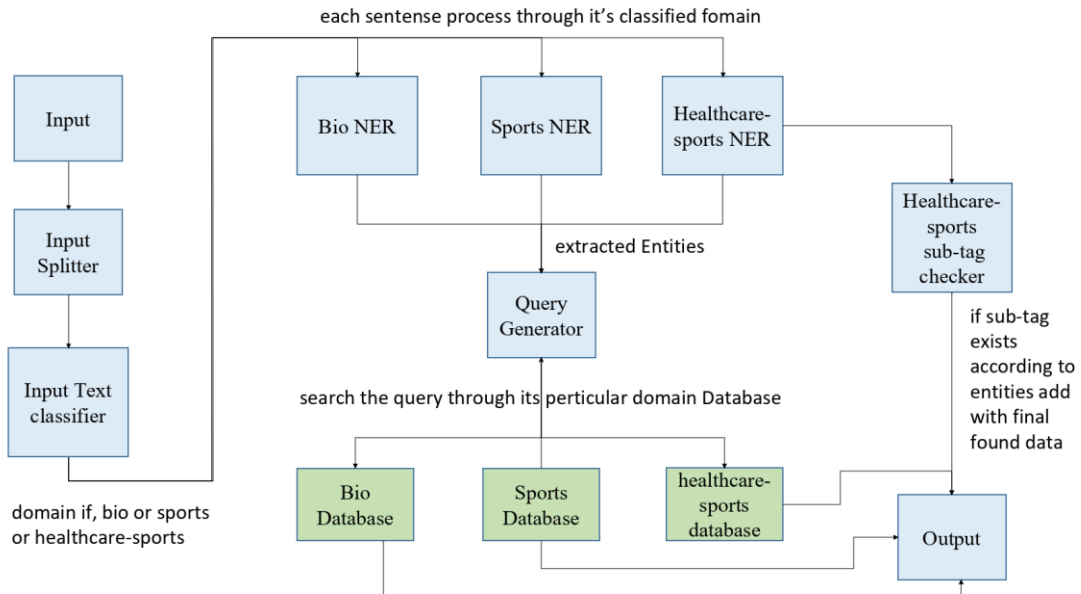


Figure 1: System Architecture

The system follows a modular NLP-based architecture designed to process unstructured user input and extract structured, domain-specific information. It integrates a sequence of intelligent modules that operate step-by-step—from text input to domain classification, named entity recognition, semantic matching, sub-tag identification, and final output generation. The architecture is crafted to support three overlapping domains : Bio, Sports, and Healthcare-Sports, enabling contextual interpretation across highly specialized and hybrid content.

Upon receiving a free-form input (such as a paragraph), the system cleans and splits the input into individual sentences to ensure independent analysis. Each sentence is then passed through a BERT-based domain classifier, which assigns it to the most appropriate domain. Based on this classification, the sentence is routed to a corresponding domain-specific NER model to extract relevant entities such as diseases, injuries, drugs, symptoms, players, and events.

The extracted entities are then used to query a domain-specific database using semantic matching logic that supports partial and case-insensitive overlaps. For sentences classified under the Healthcare-Sports domain, a specialized sub-tag detection module analyses the entities to identify specific subcategories such as *injury*, *mental health*, or *doping*. Finally, the system generates a structured output consisting of the predicted

domain, extracted entities, relevant sub-tags (if applicable), and top-matching sentences from the dataset. This architecture supports real-time information Extraction with high accuracy and contextual depth, making it suitable for intelligent text analysis in domain-blended content.

1. Input Processing and Sentence Split.

The user inputs free-form text, which can be a paragraph or multiline entry. The system performs basic cleaning and splits the text into individual sentences. This ensures that each sentence is handled independently, preserving semantic clarity and enabling precise classification and extraction.

2. Domain Classification

Each sentence is passed through a BERT-based classifier to determine whether it belongs to the *Sports*, *Bio*, or *Healthcare-Sports* domain. For instance, a sentence like “Cristiano Ronaldo complained of fatigue and was prescribed ibuprofen” would be classified under **Healthcare-Sports**, due to its mixed medical and athletic context.

3. Named Entity Recognition (NER)

Based on the assigned domain, the sentence is routed to one of the three specialized NER models :

- **Sports NER** extracts entities like *PLAYER*, *SPORT*, *TEAM*, *STAT*, etc.
- **Bio NER** detects entities like *DISEASE*, *SYMPTOM*, *DRUG*, *TEST*, etc.
- **Healthcare-Sports NER** identifies hybrid entities, including *INJURY*, *MENTAL_HEALTH*, and *DOPING_TEST*.

For example, from the sentence “Cristiano Ronaldo complained of fatigue and was prescribed ibuprofen,” the system would extract :

Cristiano Ronaldo → *PLAYER*, fatigue → *SYMPTOM*, ibuprofen → *DRUG*.

4. Semantic Matching and Extraction

Extracted entities are used to search domain-specific CSV datasets. The system performs a match based on partial string overlaps, retrieving sentences that share common entities. Matches are sorted based on the number of overlapping entities. For example, a query with “Cristiano Ronaldo, fatigue, ibuprofen” may extract :

- “Cristiano Ronaldo experienced fatigue and was given ibuprofen after training.”
- “Ibuprofen was prescribed for fatigue symptoms in football players.”

5. Healthcare-Sports Sub-Tag Checker

This module activates only for sentences classified under the Healthcare-Sports domain. It analyses ambiguous terms (e.g., “fatigue”, “testosterone”) to detect whether they relate to injuries, mental health, or doping. The system reranks entity matches accordingly and assigns a specific sub-tag if detected.

6. Output Generation

The final output includes :

- The predicted domain

- The list of extracted entities
- The detected sub-tag (if applicable)
- A ranked list of matching sentences from the relevant domain dataset

This output enables users to extract clear, structured insights from raw, mixed-domain input in an efficient and automated way.

4. Methodology

The proposed system is designed as a modular NLP pipeline that processes unstructured natural language input and converts it into structured, domain-specific information. The core workflow consists of multiple stages, each responsible for handling a specific task in the pipeline. These stages work in sequence — from raw text input to final structured output — ensuring both domain relevance and contextual accuracy.

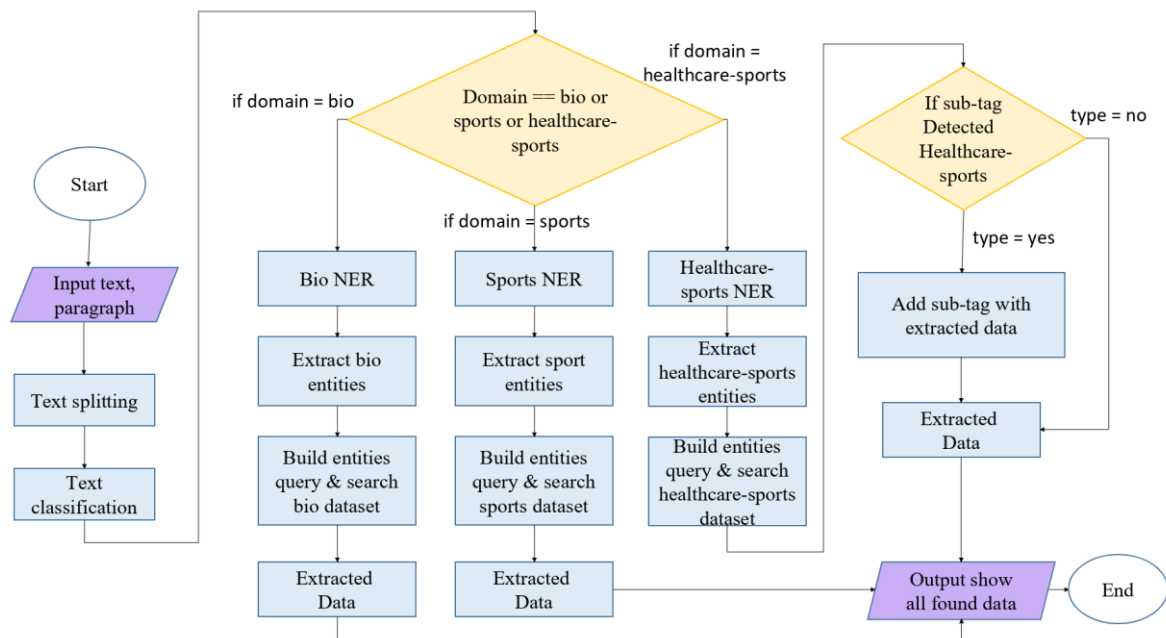


Figure 2: System Flowchart

1. Text Input

The system begins when the user provides a line or paragraph of text through the terminal interface. This input may include multiple sentences from different domains, such as clinical reports, sports commentary, or athlete-related medical events. The system is built to process and understand such mixed-domain input seamlessly.

2. Sentence Splitting

The input text is split into individual sentences using punctuation-based rules, primarily focusing on full stops (periods) as delimiters. This step ensures that each sentence is handled independently, which is essential for accurate domain classification and

downstream entity recognition. By isolating each sentence, the system prevents contextual overlap and improves precision in multi-domain environments.

3. Domain Classification

Each separated sentence is then passed through a fine-tuned BERT-based domain classifier. This classifier has been trained to assign one of three domain labels:

Bio – biomedical or clinical language

Sports – sports-only language and terminology

Healthcare-Sports – sentences containing overlap between sports and health (e.g., injuries, doping, mental health in athletes)

This classification is critical because it determines which Named Entity Recognition (NER) model will be used next. Without domain-aware classification, entity extraction would suffer from label confusion, especially in hybrid contexts.

4. Domain-Specific Named Entity Recognition (NER)

Once a sentence is classified, it is passed to the corresponding domain-specific NER model:

For Bio domain sentences, the system uses BioBERT, a specialized model pre-trained on biomedical corpora like PubMed. This model identifies entities such as DISEASE, DRUG, TEST, SYMPTOM, and more, with high accuracy.

For Sports and Healthcare-Sports domains, a general BERT-based NER model is used, fine-tuned on a custom dataset of sports and athlete-related content. It detects entities like PLAYER, TEAM, INJURY, MENTAL_HEALTH, and DOPING_TEST.

Each model outputs a list of labelled tokens in BIO format (Beginning, Inside, Outside) for precise entity mapping.

5. Sub-Tag Detection (Only for Healthcare-Sports)

If a sentence is classified under the Healthcare-Sports domain, an additional sub-tagging module is triggered. This module uses simple rule-based logic (if-else conditions) to detect deeper context based on entity types. Depending on the presence of specific entities, the sentence may be tagged with: Mental Health, Injury, Doping

For example, if entities like "anxiety" or "depression" are found, the sub-tag Mental Health is applied. This enhances result granularity and makes the system suitable for fine-grained analysis.

6. Dataset Search (Entity-Based Extraction)

The extracted entities are then used as search queries to look up related sentences in offline, pre-indexed CSV datasets, one for each domain. A custom search logic scores and ranks dataset entries based on:

- Case-insensitive string match
- Partial and full word overlaps
- Number of matched entities
- Top-matching results are retrieved for display.

7. Result Display

Finally, the system compiles and displays the results in a structured, user-readable format via the terminal. Each output includes:

- The predicted domain
- Extracted entities (with labels)
- Assigned sub-tag (if Healthcare-Sports)
- Top dataset matches sorted by relevance

This end-to-end flow allows for real-time, offline, and accurate processing of text into meaningful domain-aware insights.

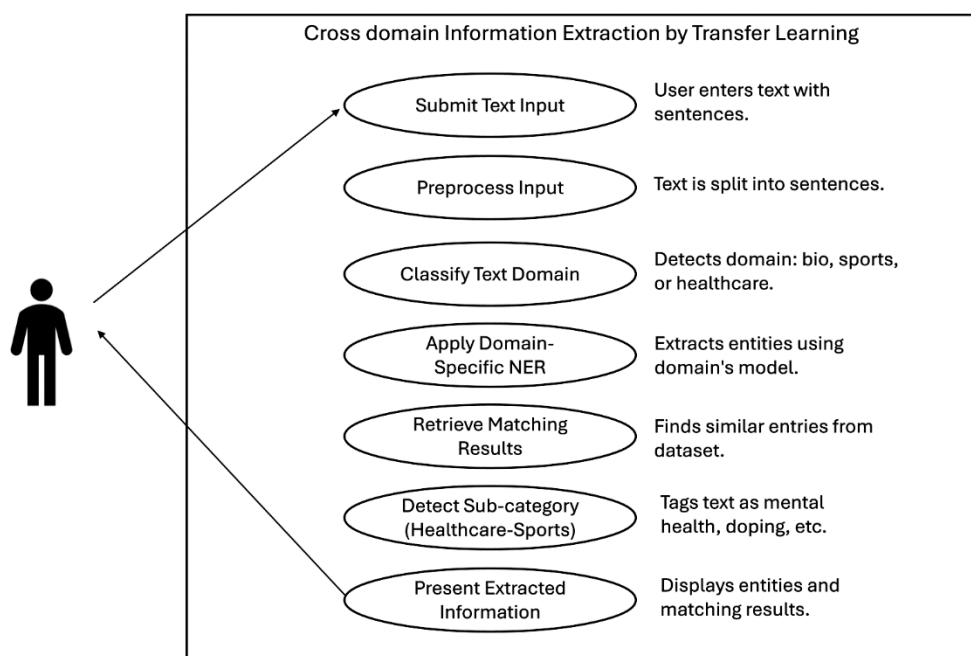


Figure 3: System Use Case Diagram

5. Domain-Specific Information

5.1 Natural Language Processing in Cross-Domain Systems

Natural Language Processing (NLP) is a foundational area within artificial intelligence that aims to bridge the gap between human communication and machine understanding. It involves a wide set of techniques for enabling machines to read, interpret, and respond to human language in a meaningful way. NLP integrates knowledge from computer science, linguistics, and machine learning to process and analyse large amounts of unstructured text data. It is widely applied in areas such as sentiment analysis, machine translation, question answering, summarization, and intelligent search systems. At the core of NLP is the ability to extract structured information from natural language, which requires models that are capable of understanding linguistic context, syntax, and semantics.

In cross-domain NLP systems, the challenge increases significantly due to the varying terminologies and contextual meanings that differ between fields. For example, a sentence about "recovery" in a financial domain might refer to economic growth, while in a medical domain, it could imply a patient's health status. Such variations require models that can adapt their understanding based on the domain in which the text occurs. Cross-domain systems must be trained or fine-tuned with domain-specific corpora to ensure their interpretations are accurate and relevant. As multiple domains often intersect—such as healthcare and sports—NLP systems must be designed to handle hybrid contexts where terminology overlaps and sentence structures become increasingly complex. Addressing this requires intelligent language models and adaptive pipelines that can not only parse the text but also assign the correct context to each phrase or entity within it.

5.2 Transformer-Based Sentence Classification using BERT

Transformer-based models have redefined the field of NLP due to their ability to model complex language dependencies and contextual relationships between words. One of the most influential architectures in this space is BERT (Bidirectional Encoder Representations from Transformers), which learns bidirectional representations by conditioning on both left and right contexts in all layers. Unlike traditional models, which process text in a single direction or rely on fixed-length vectors, BERT dynamically generates deep contextual embeddings that are sensitive to both local and global structures in language. This makes it particularly effective for downstream tasks like text classification, question answering, and language inference.

Sentence classification using BERT involves fine-tuning the pre-trained model on a labelled dataset, where each sentence is associated with a category or class. The model learns to associate linguistic patterns, word usage, and semantic cues with the target labels. In cross-domain settings, this classification is critical for determining which subsequent processing steps to take, especially when the pipeline depends on routing the input to domain-specific modules. Transformer models excel at handling ambiguity, long-range dependencies, and

subtle context differences—making them ideal for categorizing sentences into predefined classes across multiple domains. Fine-tuning BERT for sentence classification not only improves performance on specific tasks but also allows the model to adapt to the specialized language features of each domain, providing a robust and scalable classification solution.

5.3 Named Entity Recognition for Domain-Specific Entity Extraction

Named Entity Recognition (NER) is one of the most essential tasks in information extraction, and it involves identifying spans of text that correspond to predefined entity categories such as person names, organizations, locations, medical conditions, and numerical quantities. The core goal of NER is to convert unstructured text into structured representations by tagging the entities of interest. Traditional NER systems relied on rule-based approaches or simple statistical models. However, modern NER systems are primarily based on deep learning techniques, often powered by contextual embeddings generated by transformer architectures such as BERT.

Domain-specific NER extends this concept further by tailoring entity detection models to specific fields. In biomedical literature, for instance, entities of interest may include drugs, diseases, tests, and anatomical parts, while in the legal domain, the focus may shift toward contracts, regulations, or institutions. The language, formatting, and semantics in these domains differ significantly, which means generic NER systems tend to perform poorly when transferred without adaptation. To address this, domain-specific datasets are used to train NER models that understand not only what constitutes an entity but also how its form and function vary based on context. These models are usually trained with BIO (Begin, Inside, Outside) tagging schemes, which allow them to identify both single-word and multi-word entities in complex sentence structures.

Furthermore, domain-specific NER plays a crucial role in enabling intelligent search, automated indexing, and analytical processing of specialized documents. It enables machines to identify key pieces of information from large text corpora and relate them to structured knowledge bases or datasets. In complex domains, where entity types may overlap or share similar linguistic patterns, the accuracy of NER directly impacts the overall effectiveness of the system.

5.4 Entity-Based Structured Information Extraction

Entity-based structured information Extraction refers to the process of searching and extracting relevant content from a structured or semi-structured dataset using entities identified in a user's query. Traditional information Extraction systems operate on keyword matching, which often fails to capture semantic relationships or context-aware intent. In contrast, entity-based Extraction focuses on recognizing and matching core concepts, allowing for a more intelligent and targeted search experience. This is particularly valuable in professional fields where terminology is complex, and precise answers are needed—such as in healthcare, scientific research, or legal information systems.

In a typical entity-based Extraction system, input text is first analysed using Named Entity Recognition or similar techniques to extract meaningful units such as names, conditions, drugs, or players. These entities are then used to formulate a query that is matched against a curated dataset, often organized in a tabular format like CSV or a structured database. Matching techniques can include exact matching, fuzzy string matching, or semantic similarity scoring, depending on the use case. The system ranks results based on the number and relevance of overlapping entities, returning entries that closely align with the user's intent.

This approach enhances Extraction precision and relevance, especially when users are dealing with exploratory queries or need results beyond simple lexical matches. It supports intelligent recommendations, evidence-based research, and automated content linking. Moreover, entity-based Extraction can be extended to incorporate multilingual support, ontology alignment, and knowledge graph integration, making it a powerful component in advanced NLP architectures.

5.5 Importance of Domain Adaptation in NLP Tasks

Domain adaptation is the process of refining or adjusting a machine learning model to perform effectively on a specific domain that differs from the one it was originally trained on. In natural language processing, this is especially important because the structure, vocabulary, and semantics of language vary widely across different fields. A model trained on general news articles may fail to interpret biomedical literature accurately, as it would not be familiar with clinical jargon or specialized abbreviations. Domain adaptation ensures that the model not only understands the syntax of the language but also the context and meaning as used in a particular industry or discipline.

There are various techniques used in domain adaptation, including supervised fine-tuning, where a pre-trained model like BERT is trained further on a labelled dataset from the target domain. Semi-supervised learning and transfer learning also play key roles in adapting to domains with limited annotated data. These methods help the model retain its general language understanding while specializing in domain-specific patterns. Adaptation improves outcomes in tasks such as classification, entity recognition, sentiment analysis, and question answering.

The importance of domain adaptation lies in its ability to improve accuracy, reduce misinterpretation, and enhance the overall usefulness of NLP systems in real-world applications. As AI systems become more integrated into critical sectors such as healthcare, finance, and law, the need for reliable and specialized language models becomes essential. Domain adaptation bridges the gap between general-purpose AI and high-stakes, domain-specific use cases, enabling models to function with expert-level understanding in their respective areas.

6. Hardware and Software

6.1 Hardware

- Processor :
Minimum : 2 GHz dual-core CPU
Recommended : Quad-core CPU (e.g., Intel i5/i7 or AMD Ryzen 5/7) to ensure smooth processing of NLP tasks and transformer model execution.
- Memory (RAM) :
Minimum : 8 GB
Recommended : 16 GB or more, especially for handling multiple models or working with medium-to-large datasets.
- Graphics Card (GPU) :
Recommended : NVIDIA GPU with at least 4 GB VRAM (e.g., GTX 1650 or higher) for BERT model fine-tuning and accelerated training.
Alternative : Google Colab or Kaggle for free cloud-based GPU/TPU access.
- Storage :
Minimum 20 GB of available space is required to store datasets, model checkpoints, temporary logs, and result outputs.
SSD is preferred over HDD for faster data access and file operations during processing.

6.2 Software

- Operating System :
Ubuntu 20.04 / 22.04 LTS or Windows 10 / 11 (64-bit versions)
- Programming Language :
Python 3.7 or higher (Python 3.10 recommended)
- Core Libraries and Frameworks :
 - transformers (Hugging Face) – for loading and fine-tuning BERT models
 - torch (PyTorch) – for model training and inference
 - scikit-learn – for performance evaluation
 - pandas, numpy – for data manipulation and analysis
 - nltk or spaCy – for text pre-processing and tokenization (optional)
- IDE / Editor :
Visual Studio Code, PyCharm, or Jupyter Notebook for development, experimentation, and debugging.
- Dataset Format :
.csv files used for structured dataset storage (domain classification data and NER-tagged data)
- Cloud Platforms (optional) :
Google Colab / Kaggle for GPU-based execution if local hardware is limited.

7. Implementation Details

7.1 Technology Used

This project is developed using a modular and scalable stack of open-source tools focused on Natural Language Processing (NLP). The entire system is implemented in Python 3.10, taking advantage of its robust ecosystem for deep learning, data processing, and model evaluation.

The core of the system relies on the Hugging Face Transformers library, specifically using pre-trained BERT models for both sentence classification and token-level Named Entity Recognition (NER). For fine-tuning and custom training, PyTorch serves as the backend deep learning framework, allowing the construction of classification and token prediction heads on top of BERT encoders.

For model evaluation, scikit-learn is used to compute metrics such as accuracy, precision, recall, and F1-score. Data processing and manipulation are carried out using Pandas and NumPy, while CSV files are used as the storage format for both domain classification datasets and entity-tagged corpora.

Development is carried out in Jupyter Notebook and Google Colab, which provide interactive environments for testing, debugging, and visualizing outputs. Cloud execution via Colab also allows access to GPU acceleration during model training, making the solution hardware-independent. The system remains light-weight and does not rely on any server-side deployment or database. All results, entity mappings, and queries are handled locally using file-based operations.

7.2 System Pipeline Architecture

The system follows a multi-stage pipeline architecture designed to handle unstructured user input, process it sentence by sentence, and extract domain-specific information for downstream Extraction. Each stage is modular, allowing independent training, debugging, and improvement of each component.

The pipeline begins with the Input Processor, which accepts either a single line or a multiline paragraph. The input is split into individual sentences using basic punctuation-based segmentation to ensure each unit is handled separately. This avoids context leakage and ensures each sentence is independently classified and analysed.

The next module is the Domain Classifier, a fine-tuned BERT model trained to predict whether a sentence belongs to the Bio, Sports, or Healthcare-Sports domain. This classification determines the flow of the sentence through the rest of the system. For instance, a sentence discussing a disease and a player will be directed to the healthcare-sports pipeline rather than treated as purely biomedical or sports-related.

Once classified, the sentence is routed to its respective Named Entity Recognition (NER) model. Each model is trained on BIO-tagged datasets customized for its domain. The Bio NER detects entities such as DISEASE, DRUG, SYMPTOM, and TEST; the Sports NER identifies PLAYER, TEAM, SPORT, and STAT; while the Healthcare-Sports NER integrates both sets and introduces additional entities like MENTAL_HEALTH, INJURY, and DOPING_TEST.

In the case of healthcare-sports inputs, an additional Sub-Tag Detector analyses extracted entities and assigns context-based subcategories like MENTAL_HEALTH, DOPING, or INJURY using predefined heuristics based on entity co-occurrence and type combinations.

Finally, the Entity Query Engine takes the recognized entities and performs a similarity-based search across structured CSV datasets. The Extraction module calculates a matching score based on partial string overlap and entity count, returning the top 5 most relevant sentences as evidence. This results in a fully functional NLP pipeline that handles text classification, information extraction, and Extraction without needing an external database or API.

7.3 Algorithms Used

Several algorithms are employed at each stage of the system to ensure high accuracy and modularity.

7.3.1. Sentence Classification Algorithm

- Model: BERT-base uncased
- Objective: Multi-class sentence classification
- Fine-Tuning: On 3-class dataset (Bio, Sports, Healthcare-Sports)
- Loss Function: CrossEntropyLoss
- Output: Class label corresponding to sentence domain
- Metric: Accuracy, Precision, Recall, F1-score

7.3.2. Token-Level Named Entity Recognition

- Model: Domain-specific BERT models with a token classification head
- Training Format: BIO-tagged datasets
- Loss Function: CrossEntropyLoss applied token-wise
- Output: List of entity spans with types
- Metric: Precision, Recall, and F1-score (entity-level)

7.3.3. Sub-Tag Detection Heuristic (Healthcare-Sports Only)

- Detect subcategories like MENTAL_HEALTH, INJURY, and DOPING
- Rule: Based on presence of certain entity types (e.g., MENTAL_HEALTH terms → MENTAL_HEALTH tag)
- Uses co-occurrence frequency and tag combinations to infer sub-context

7.3.4. Entity Matching and Extraction Algorithm

- Input: Extracted entities from NER
- Dataset: CSV file with fields text and all_entities
- Logic:
 - Normalize all entities (lowercase, stripped)
 - Compare with each dataset entry using partial string and exact matches
 - Assign score = number of matched entities
- Output: Top 5 results sorted by score
- Use Case: Evidence-based Extraction with no SQL/database dependency.

8. Result Analysis

Stage 1: Sentence Classification – BERT-Based Domain Classifier.

In the first stage of the system pipeline, the user provides a line or paragraph of free-form natural language input. The system first splits the input text into individual sentences using the period (.) as a separator. This sentence-level separation is necessary because a single input may contain multiple thoughts or statements, each belonging to a different domain. Splitting ensures that each sentence is evaluated and classified independently.

After splitting, each sentence is passed to a fine-tuned BERT-base model, which acts as a multi-class classifier. This model has been trained to predict the domain of a sentence as one of the following:

- Bio – for biomedical and clinical statements
- Sports – for content related to games, players, scores, or events
- Healthcare-Sports – for sentences that mix health and sports topics (e.g., injuries, doping, athlete mental health)

.

```
PS C:\Users\kaustubh chaudhari\OneDrive\Desktop\ProjectCode> python main.py
Enter your text (single or multiple lines):

cancer was detected by x-ray. Rohit sharma tested positive after consumption of banned substance after saliva test. Sachin scored century in last match at wankdede

Split into sentences, classified domains, and extracted entities with search results:

1. cancer was detected by x-ray → 📁 Domain: bio
Asking to truncate to max_length but no maximum length is provided and the model has no predefined maximum length. Default to no truncation.

2. Rohit sharma tested positive after consumption of banned substance after saliva test → 📁 Domain: healthcare-sports

3. Sachin scored century in last match at wankdede → 📁 Domain: sports
PS C:\Users\kaustubh chaudhari\OneDrive\Desktop\ProjectCode>
```

Figure 4: Result Stage 1 Classification

Example 1 (Multi-sentence Input):

Input: "The patient was prescribed paracetamol. LeBron James suffered an ankle injury. Virat Kohli hit a century last night."

Classification Output:

- "The patient was prescribed paracetamol." → Bio
- "LeBron James suffered an ankle injury." → Healthcare-Sports

- “Virat Kohli hit a century last night.” → Sports

Example 2 (Single complex sentence):

Input: "After testing positive for a banned substance, the athlete was suspended."

Classification Output:

- Sentence: "After testing positive for a banned substance, the athlete was suspended." → Healthcare-Sports

Example 3 (Medical content):

Input: "MRI reports showed signs of brain hemorrhage."

Classification Output:

- Sentence: "MRI reports showed signs of brain hemorrhage." → Bio

Classification Results:

The domain classifier performed well on a combined dataset, with accuracy of 90.67% precision of 91.58%, recall of 90.67%, and F1-score of 90.61%.

Domain	Precision	Recall	F1-Score
bio	0.9259	1.0000	0.9615
healthcare-sports	0.8125	1.0000	0.8966
sports	1.0000	0.8000	0.8889

Figure 5: Classification Results

Stage 2: Named Entity Recognition – Domain-Specific Token Classification

Once each sentence has been classified into its respective domain, the system moves into the Named Entity Recognition (NER) stage. In this step, the classified sentence is sent to a domain-specific BERT-based token classification model that extracts relevant entities from the sentence using the BIO tagging format. Each domain has its own fine-tuned model trained on a specific set of entity types relevant to its context.

The NER model reads each token in the sentence and assigns it a label such as:

- B-ENTITY_TYPE → Beginning of an entity
- I-ENTITY_TYPE → Inside of the same entity
- O → Outside of any entity

```
PS C:\Users\kaustubh_chaudhari\OneDrive\Desktop\ProjectCode> python main.py
Enter your text (single or multiple lines):

The patient was diagnosed with cancer after a blood test.Virat Kohli complained of stress after the match.Sachin scored a century in the final match.

Split into sentences, classified domains, and extracted entities with search results:

1. The patient was diagnosed with cancer after a blood test → Domain: bio
Asking to truncate to max_length but no maximum length is provided and the model has no predefined maximum length. Default to no truncation.
{'cancer': 'B-DISEASE', 'blood': 'B-TEST', 'test': 'I-TEST'}

2. Virat Kohli complained of stress after the match → Domain: healthcare-sports
{'Virat': 'B-SPORT_CONTEXT', 'Kohli': 'I-SPORT_CONTEXT', 'stress': 'B-MENTAL_HEALTH'}

3. Sachin scored a century in the final match → Domain: sports
{'Sachin': 'B-PLAYER', 'century': 'B-STAT', 'final': 'B-EVENT', 'match': 'I-EVENT'}
PS C:\Users\kaustubh_chaudhari\OneDrive\Desktop\ProjectCode>
```

Figure 6: Result Stage 2 NER Extracion

Bio Domain – BioBERT NER Model

- Model: BioBERT (a BERT variant pre-trained on biomedical literature)
- Purpose: To detect medical terms such as diseases, drugs, tests, anatomy, dosage, etc.
- Example Entities:
 - DISEASE, DRUG, SYMPTOM, ANATOMY, TEST, DOSAGE
 -

Example 1 (Bio):

Sentence: "The patient was prescribed 500 mg of paracetamol for fever."

NER Output:

- "paracetamol" → B-DRUG
- "500 mg" → B-DOSAGE

- “fever” → B-SYMPTOM

Sports Domain – General BERT NER (Sports-trained)

- Model: General BERT, fine-tuned on sports-specific labelled data
- Purpose: To extract information related to players, events, stats, awards, etc.
- Example Entities:
 - PLAYER, TEAM, SPORT, STAT, EVENT, VENUE, AWARD

Example 2 (Sports):

Sentence: "Cristiano Ronaldo scored a hat-trick in the World Cup final."

NER Output:

- “Cristiano Ronaldo” → B-PLAYER I-PLAYER
- “hat-trick” → B-STAT
- “World Cup” → B-EVENT

Healthcare-Sports Domain – Hybrid BERT NER Model

- Model: Custom BERT trained on a combined entity set from both Bio and Sports, with additional healthcare-sports tags
- Purpose: To handle overlapping terms and hybrid concepts
- Additional Entities:
 - INJURY, MENTAL_HEALTH, DOPING_TEST, SPORT_CONTEXT

Example 3 (Healthcare-Sports):

Sentence: "LeBron James suffered from depression after the doping test."

NER Output:

- “LeBron James” → B-SPORT_CONTEXT I-SPORT_CONTEXT
- “depression” → B-MENTAL_HEALTH
- “doping test” → B-DOPING_TEST I-DOPING_TEST

Example 4 (Another Hybrid):

Sentence: "The athlete experienced a torn ligament and was given ibuprofen."

NER Output:

- “torn ligament” → B-INJURY I-INJURY
- “ibuprofen” → B-DRUG

Stage 3. Information Extraction and Healthcare-sports labelling

After domain classification and entity recognition are completed, the final stage of the pipeline focuses on information extraction, where the system uses the recognized entities to search through structured domain-specific datasets. Additionally, for sentences classified under Healthcare-Sports, the system performs sub-category labelling to further identify whether the content relates to specific concerns such as injury, mental health, or doping. This final stage makes the output useful, contextual, and actionable.

Dataset-Based Information Extraction

- Purpose: To find real examples or evidence from curated .csv datasets based on extracted entities.
- Process:
 - Each domain (Bio, Sports, Healthcare-Sports) has its own dataset in CSV format.
 - The extracted entities from NER are matched against the all_entities column of each CSV entry.
 - A scoring system is used to rank results:
 - Higher scores for full or partial matches
 - Case-insensitive and substring matching allowed
 - Top N results (usually 2–5) are returned for each sentence.

```
PS C:\Users\kaustubh_chaudhari\OneDrive\Desktop\ProjectCode> python main.py
Enter your text (single or multiple lines):

The patient was diagnosed with cancer after a blood test.Virat Kohli complained of stress after the match.Messi was the top scorer in the recent tournament..

Split into sentences, classified domains, and extracted entities with search results:

1. The patient was diagnosed with cancer after a blood test → 📁 Domain: bio
Asking to truncate to max_length but no maximum length is provided and the model has no predefined maximum length. Default to no truncation.
{'cancer': 'B-DISEASE', 'blood': 'B-TEST', 'test': 'I-TEST'}
🔍 Top Matching Results from Dataset:
1. cough syrup was prescribed after blood test confirmed cancer.
2. Doctors advised blood test for cancer.

2. Virat Kohli complained of stress after the match → 📁 Domain: healthcare-sports
{'Virat': 'B-SPORT_CONTEXT', 'Kohli': 'I-SPORT_CONTEXT', 'stress': 'B-MENTAL_HEALTH'}
This sentence is related to mental health.
🔍 Top Matching Results from Dataset:
1. Virat Kohli skipped practice due to ongoing stress disorder.
2. Virat Kohli spoke publicly about suffering from stress disorder.

3. Messi was the top scorer in the recent tournament → 📁 Domain: sports
{'Messi': 'I-PLAYER', 'scorer': 'I-STAT', 'tournament': 'B-EVENT'}
🔍 Top Matching Results from Dataset:
1. Lionel Messi and Cristiano Ronaldo were the highlights of the tournament.
2. Lionel Messi's brilliance secured PSG a place in the tournament.
PS C:\Users\kaustubh_chaudhari\OneDrive\Desktop\ProjectCode>
```

Figure 7 : Result Stage 3 Extraction and Sub-tag Detection

Example 1 (Bio Extraction):

Input Sentence: "The patient was prescribed paracetamol for fever."

Extracted Entities: paracetamol, fever

Search Result:

- "Paracetamol is commonly used to treat fever in viral infections."
- "Doctors often prescribe paracetamol for high fever cases."

Example 2 (Sports Extraction):

Input Sentence: "Rohit Sharma hit a double century at Eden Gardens."

Extracted Entities: Rohit Sharma, double century, Eden Gardens

Search Result:

- "Rohit Sharma scored 200+ runs at Eden Gardens in 2014."
- "Double centuries in ODIs are rare; Sharma holds multiple records."

Healthcare-Sports Sub-Category Labelling

For sentences identified as Healthcare-Sports, the system performs additional tagging based on the types of entities detected. This is done using a rule-based logic module.

- Entity Types Triggering Sub-Tags:
 - MENTAL_HEALTH → Triggers Mental Health sub-tag
 - INJURY → Triggers Injury sub-tag
 - DOPING_TEST, DRUG in sports context → Triggers Doping sub-tag
- This tagging helps flag sensitive or critical issues in hybrid health-sport contexts.

Example 3 (healthcare-sports extraction)

Input Sentence:

"Neeraj Chopra suffered from a torn ligament and took painkillers."

Extracted Entities: Neeraj, Chopra, torn, ligament, painkillers

Sub-Tag Assigned:

- Injury

Retrieved Results:

- "Neeraj Chopra missed training due to ligament injury."

- “Athletes with torn ligaments are often prescribed painkillers for recovery.”
- “Painkillers were used to manage ligament pain in javelin throwers.”

Over All Results:

Model	Accuracy	Precision	Recall	F1-Score
Bio NER	0.9073	0.7500	0.7333	0.7416
Healthcare NER	0.9602	0.9600	0.9231	0.9412
Sports NER	0.8696	0.6667	0.7907	0.7234
Domain Classifier	0.9067	0.9158	0.9067	0.9061

Figure 8: Over All Models Performance Results

9. Conclusion

This project demonstrates the successful development of a modular and intelligent information extraction system capable of handling mixed-domain textual data. By combining domain classification, named entity recognition (NER), and contextual sub-tagging, the system transforms raw user input into structured and meaningful output. Each stage of the pipeline—from sentence splitting to dataset-based result Extraction—is carefully designed to operate independently yet cohesively, ensuring high accuracy and adaptability across domains.

The incorporation of BERT-based models for sentence-level classification enables the system to robustly identify the correct domain (Bio, Sports, or Healthcare-Sports) for each sentence, even in complex or ambiguous cases. This is followed by domain-specific NER modules, including the use of BioBERT for medical content and custom-trained BERT variants for sports-related inputs. These models are fine-tuned on annotated datasets using the BIO tagging format to achieve precise and context-aware entity extraction.

What sets this system apart is its ability to handle hybrid domains like Healthcare-Sports. In this area, where standard models often fail due to overlapping medical and athletic terminology, the system applies an additional rule-based sub-tagging logic to identify critical contexts such as Mental Health, Injury, and Doping. This granular tagging allows users to not only extract data but also understand the deeper context behind each sentence.

In conclusion, the system effectively bridges the gap between unstructured user input and structured, domain-relevant information. It is particularly suited for applications involving biomedical text mining, sports analytics, and health-sports research. With further improvements in dataset size, model tuning, and UI enhancement, this project can be extended into a more interactive and scalable solution for real-world deployment in academic, clinical, or media analysis settings.

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